

An Edge-Based Computational Intelligence Framework for Vision-Driven Cattle Health Monitoring in Smart Agriculture

Abstract—Animal health monitoring remains a major challenge in the Indian dairy industry, where most of the small- and marginal-level dairy farmers are using manual monitoring techniques to assess the health of the cattle population. Most existing automated livestock monitoring systems rely heavily on wearable sensors and cloud-connected infrastructure. This work presents an edge-based computational intelligence framework for vision-driven cattle health monitoring using IoT-enabled infrastructure. The proposed system performs deep learning-based visual intelligence, including cattle detection, persistent identity association and multi-class behavior recognition, through continuous video analysis at the edge. All computational intelligence-based inference and decision-making are executed locally on the edge device, with optional cloud connectivity limited to data synchronization and model updates. The framework is deployed on an NVIDIA Jetson Orin Nano platform, enabling cost-effective and connectivity-independent operation in real dairy farm settings. Extensive evaluation using real-world farm video data demonstrated strong edge-level performance, achieving a detection mAP of 95.2%, identity association accuracy of 96.0% and overall behavior classification accuracy of 98.21% with a macro-average F1-score of 97.72%. The results validate the effectiveness of the proposed computational intelligence-driven edge vision framework for real-time cattle monitoring and preventive livestock healthcare in smart agriculture.

Index Terms—Computational Intelligence, Deep Learning, Edge AI, Computer Vision, Vision-Based Cattle Monitoring, Behavior Recognition, Precision Livestock Farming, Smart Agriculture

I. INTRODUCTION

India hosts one of the largest cattle populations globally, with dairy farming forming a crucial component of rural livelihoods and the national economy [1]. However, cattle health and behavioural monitoring in many rural and semi-urban dairy farms still relies heavily on manual inspection and subjective visual observations, making early detection of diseases, behavioural abnormalities and welfare-related issues difficult, labour-intensive and inconsistent [2]. Delayed identification of abnormal activities such as lameness, reduced feeding behaviour, prolonged inactivity and heat stress can negatively affect milk productivity, animal welfare and overall farm profitability.

Recent advancements in deep learning, computer vision and precision livestock farming have enabled automated livestock monitoring through object detection, multi-object tracking, behavioural analysis and cattle re-identification techniques [3], [4]. Vision-based systems integrated with convolutional neural networks and intelligent computer vision models have

demonstrated promising performance in cattle identification, lameness analysis, feeding monitoring, posture estimation and behavioural classification tasks under controlled and semi-controlled farm environments [5]–[7]. Furthermore, skeleton-based activity recognition approaches and temporal deep learning frameworks have shown improved capability for recognising cattle movement and behavioural patterns from continuous video streams [8], [9].

Persistent identity-aware livestock monitoring has gained increasing attention in recent years. Multi-feature cattle identification techniques based on facial characteristics, muzzle patterns and ear-tag information have enabled more reliable long-term monitoring of individual animals within dense herd environments [10]. Simultaneously, recent surveys on YOLO-based agricultural object detection and lightweight edge-oriented inference systems indicate that optimized deep learning architectures are increasingly suitable for real-time deployment in resource-constrained agricultural environments [11], [12]. Deep learning approaches have additionally been explored for disease-related applications such as mastitis detection and teat localization, further highlighting the growing importance of intelligent vision systems in precision livestock healthcare management [13], [14].

Although embedded AI systems and TinyML-based livestock sensing platforms have demonstrated promising feasibility for low-power cattle behaviour monitoring [15], developing robust and scalable vision-driven intelligence systems for operational dairy farm deployment remains challenging. Existing systems are often constrained by limited behavioural scope, lack of persistent identity association, sensitivity to illumination variation and occlusion, dependence on controlled environments, or insufficient validation under real-world farm settings [16].

In this work, we present an IoT-enabled edge-based intelligent vision framework for continuous cattle health monitoring tailored to small and marginal dairy farms. Unlike conventional livestock monitoring systems that independently perform detection, tracking or behaviour classification tasks, the proposed framework integrates YOLOv8-based cattle detection, persistent multi-object tracking, OCR-driven identity association, temporal behaviour reasoning, adaptive threshold-based inference and multi-class behaviour recognition within a unified real-time edge inference pipeline.

A key contribution of the framework is the incorporation of persistent identity-aware monitoring through ear-tag detection

and OCR-based identity inference, enabling continuous tracking and long-term behavioural analysis of individual cattle. Furthermore, the proposed system incorporates asynchronous multi-threaded execution, queue-based inference scheduling, sliding-window temporal smoothing and concurrent multi-camera processing to achieve stable low-latency inference on resource-constrained NVIDIA Jetson Orin Nano hardware.

The proposed framework is experimentally validated using real-world datasets collected through continuous field deployment in operational dairy farms involving illumination variation, partial occlusion, dense cattle presence and long-duration monitoring. Unlike many existing studies that focus on isolated monitoring tasks under controlled environments, the proposed work demonstrates the practical integration and deployment of multiple intelligent vision components within a unified identity-aware livestock monitoring architecture suitable for resource-constrained smart agriculture environments.

II. RELATED WORK

Deep learning-based computer vision techniques have been widely investigated for automated cattle behavior monitoring in precision livestock farming. In [17], a computer vision-based lameness detection framework was proposed using image processing and behavioural motion analysis under real-world farm environments. Although the framework demonstrated promising capability for automated cattle health assessment, it primarily focused on lameness detection and did not address persistent identity association or long-term behavioural aggregation.

Machine vision techniques for cattle behaviour analysis have also been extensively explored in livestock monitoring research. A comprehensive review presented in [18] discussed feeding analysis, locomotion assessment, posture recognition and activity monitoring using deep learning and computer vision approaches. The study additionally highlighted deployment challenges including illumination variation, occlusion and scalability limitations in dense farm environments. However, most reviewed systems focused on isolated behavioural tasks without unified identity-aware monitoring.

Precision livestock farming technologies integrated with intelligent sensing and welfare-oriented monitoring systems were further investigated in [19]. The work emphasized automated livestock monitoring for continuous animal welfare assessment but did not address real-time edge-based multi-object behaviour inference or persistent cattle re-identification.

In [20], a foundational overview of precision livestock farming technologies involving automated sensing, behavioural monitoring and intelligent farm management systems was presented. Although the study highlighted the importance of continuous livestock monitoring in smart agriculture environments, practical challenges associated with low-latency edge inference and integrated real-time health assessment remained insufficiently addressed.

Overall, existing livestock monitoring studies primarily focus on isolated tasks such as behaviour classification, activity

recognition or welfare assessment under controlled or cloud-dependent settings. Persistent cattle identity association, long-term behavioural analysis and integrated health inference remain underexplored, particularly within low-cost edge-based intelligent vision frameworks validated under real-world dairy farm environments.

III. MATERIALS AND METHODS

A. Materials

1) *Dataset Collection*: The dataset was collected using commercially available high-definition USB cameras (Zebtronics FHD USB Webcam) equipped with a 2.0 MP CMOS sensor capable of recording video at 30 frames per second. Video data were captured inside cattle sheds at one organized goshala and two small-scale dairy farms located in Doddaballapur, Bengaluru Rural District, Karnataka, India. The recorded videos had a resolution of 1920×1080 pixels and were subsequently processed to extract individual frames for analysis. All frames were manually annotated to support supervised learning of the deep learning-based models.

While the dataset is collected from specific farm locations in Karnataka, it captures realistic farm conditions such as occlusion, illumination variation and dense cattle presence. Future work will focus on expanding data collection across diverse geographical regions to improve model generalization and robustness.

The dataset comprises five behavior classes relevant to cattle health assessment, namely drinking, eating, lying, ruminating and standing. The annotated dataset was partitioned into training, validation and test sets, as summarized in Table I.

TABLE I
DISTRIBUTION OF CATTLE BEHAVIOR SAMPLES ACROSS TRAINING, VALIDATION AND TEST SETS

Behavior	Train	Validation	Test
Drinking	1948	556	280
Eating	3026	864	433
Lying	2090	597	300
Ruminating	1240	354	178
Standing	6137	1753	878
Total	14441	4124	2069

2) *Handling Class Imbalance*: The observed class imbalance reflects the natural behavioral distribution of cattle in real farm environments. However, rather than utilizing resampling methods, class normalization was followed during training to ensure equal usage and relevance of all classes of cattle behavior. Class-wise precision, recall and F1-score were used for performance evaluation, which are key indicators for measuring the efficacy of deep learning-based behavior recognition models used in intelligent livestock monitoring systems.

3) *Training Hardware*: Model training, evaluation and edge-level inference were performed on the NVIDIA Jetson Orin Nano 8GB embedded AI platform equipped with a 1024-core NVIDIA Ampere GPU with 32 Tensor Cores, a 6-core Arm Cortex-A78AE CPU and 8 GB LPDDR5 memory. The

platform supports CUDA-accelerated deep learning inference and asynchronous multi-threaded execution, enabling efficient deployment of YOLOv8, EfficientNetV2-S and OCR-based deep learning models for real-time edge-level cattle monitoring under practical agricultural environments.

4) *Software Environment*: The proposed framework was implemented using Python with PyTorch, OpenCV, Ultralytics YOLO, TIMM, PaddleOCR and Albumentations libraries. CUDA-enabled GPU acceleration was utilized for real-time inference and edge-level deployment optimization on the NVIDIA Jetson Orin Nano platform.

B. Computational Intelligence-Driven Perception and Reasoning Methods

1) Deep Learning-Based Cattle Detection and Tracking:

The YOLOv8 model was fine-tuned using transfer learning from pretrained COCO weights to improve convergence and detection robustness under real-world farm environments. The detector processes video frames at a resolution of 640×480 pixels and was configured to detect only cattle instances to reduce computational overhead. Detection confidence was experimentally fixed at 0.30 to balance precision and recall under practical farm conditions including illumination variation, occlusion and dense cattle presence.

Persistent cattle tracking across consecutive frames is achieved using the integrated YOLO tracking framework with temporal identity persistence enabled. Each detected cattle instance was assigned a temporary track identifier for continuous monitoring.

The detection output for a given frame is represented as

$$D(\hat{F}) = \{(b_i, c_i)\}_{i=1}^N \quad (1)$$

where b_i denotes the bounding box and c_i represents the confidence score of the detected cattle instance.

2) *Ear-Tag Detection and Identity Inference*: Persistent cattle identity association is achieved using a custom YOLO-based ear-tag detector combined with PaddleOCR-based numeric identity extraction. Ear-tag detection is performed with a confidence threshold of 0.35. To improve OCR robustness under practical farm conditions, image enhancement techniques including grayscale conversion and CLAHE-based contrast enhancement are applied before OCR inference.

The extracted identity values are mapped to temporary tracking identifiers, enabling long-term identity-aware cattle monitoring.

The OCR-based identity inference is defined as

$$ID_j = OCR(\phi(b_j)) \quad (2)$$

where $\phi(\cdot)$ denotes the ear-tag region extraction function.

3) *EfficientNetV2-S-Based Behavior Recognition*: Cattle behavior recognition was implemented using the EfficientNetV2-S deep convolutional neural network architecture with ImageNet-pretrained weights. The model was trained to classify five behavior categories: standing, lying, eating, drinking and ruminating.

Cattle regions of interest were automatically extracted using the YOLO detector and resized to 256×256 pixels during preprocessing, while the final training input size was fixed at 384×384 pixels. To improve model robustness and generalization, augmentation techniques including horizontal flipping, rotation, brightness and contrast adjustment, Gaussian noise injection and random scaling are applied using the Albumentations framework.

The model was trained using weighted categorical cross-entropy loss with the Adam optimizer, learning rate of 1×10^{-4} , batch size of 16 and ReduceLROnPlateau scheduler for 10 epochs on the NVIDIA Jetson Orin Nano platform with CUDA acceleration enabled.

The behavior prediction is obtained as

$$\hat{b}_j = \arg \max_k f_\theta(X_j) \quad (3)$$

where $f_\theta(\cdot)$ denotes the trained EfficientNetV2-S model.

To improve temporal consistency during inference, majority-voting temporal smoothing was applied over a seven-frame sliding window:

$$\tilde{b}_j^t = \text{mode} \left\{ \hat{b}_j^{t-W+1}, \dots, \hat{b}_j^t \right\} \quad (4)$$

where W denotes the temporal smoothing window length.

4) *Hyperparameter Configuration and Edge Optimization*: The major architectural configurations and hyperparameters used in the proposed framework are summarized in Table II.

TABLE II
HYPERPARAMETER CONFIGURATION

Parameter	Value
Detection model	YOLOv8
Behavior model	EfficientNetV2-S
OCR engine	PaddleOCR
Input resolution	640×480
Behavior input	384×384
ROI size	256×256
Batch size	16
Optimizer	Adam
Learning rate	1×10^{-4}
Epochs	10
Temporal window	7 frames
Detection threshold	0.30
Ear-tag threshold	0.35
Hardware	Jetson Orin Nano
GPU support	CUDA enabled
Frameworks	PyTorch, TIMM, OpenCV, Albumentations

To support low-latency edge deployment, the framework adopts lightweight deep learning architectures, GPU-accelerated inference and asynchronous multi-threaded execution. Independent capture, inference and visualization threads communicate through queue-based buffering, enabling stable multi-camera real-time processing with reduced computational overhead.

All inference and decision-making operations are executed locally on the edge device, improving scalability, reducing network overhead and enabling reliable deployment in rural smart agriculture environments.

IV. IMPLEMENTATION

A. Experimental Setup

1) *Edge-Based Video Stream Ingestion and Intelligent Preprocessing*: Video streams from multiple cameras are processed locally at the edge using a multi-threaded pipeline to support real-time inference. To balance computational efficiency and temporal coverage, frames are sampled at fixed intervals and undergo preprocessing operations, including resizing, normalization and region-of-interest (ROI) extraction.

Let the set of video streams captured from M cameras be defined as

$$\mathcal{V} = \{V_1, V_2, \dots, V_M\}, \quad (5)$$

where each stream consists of a sequence of frames

$$V_m = \{F_m^t \mid t = 1, 2, \dots, T_m\}. \quad (6)$$

Frame sampling at a fixed interval k is defined as

$$F_m^{t,s} = F_m^t \quad \text{such that } t \bmod k = 0. \quad (7)$$

Each sampled frame is preprocessed as

$$\hat{F}_m^t = \mathcal{P}(F_m^{t,s}), \quad (8)$$

where $\mathcal{P}(\cdot)$ denotes resizing, normalization and ROI extraction.

2) *Learning-Based Cattle Detection and Multi-Object Tracking*: Cattle detection and tracking are performed using a YOLOv8-based edge inference pipeline optimized for practical farm environments involving illumination variation, cattle overlap and dense multi-cattle scenes.

The detection output for a given frame is defined as

$$\mathcal{D}(\hat{F}) = \{(b_i, c_i)\}_{i=1}^N, \quad (9)$$

where b_i denotes the bounding box of the i -th detected cattle instance and c_i represents the corresponding detection confidence.

3) *Persistent Identity Inference via Edge-Based OCR*: Persistent cattle identity association is achieved using an identity inference pipeline that integrates ear-tag detection with optical character recognition (OCR). Ear-tag regions are extracted from the detected cattle instances and numeric identity information is obtained as

$$ID_j = \text{OCR}(\phi(b_j)), \quad (10)$$

where $\phi(\cdot)$ denotes the ear-tag region extraction function. The extracted identities are used to link temporary tracking identifiers with permanent cattle identities, enabling identity-aware behavioral analysis over extended time periods.

4) *Behavior Recognition and Temporal Intelligence Modeling*: Cattle behavior recognition is performed using a CNN-based multi-class deep learning model operating on cropped cattle regions. The predicted behavior label is obtained as

$$\hat{b}_j = \arg \max_k f_\theta(X_j), \quad (11)$$

where $f_\theta(\cdot)$ denotes the trained behavior recognition model.

The confidence associated with the predicted behavior is computed as

$$c_j = \max_k f_\theta(X_j)_k. \quad (12)$$

To obtain stable and robust behavioral states, temporal intelligence modeling is applied using majority voting over a sliding window of length W :

$$\tilde{b}_j^t = \text{mode}\{\hat{b}_j^{t-W+1}, \dots, \hat{b}_j^t\}. \quad (13)$$

5) *Event-Driven Intelligent Alert Generation*: An event-driven alert engine performs intelligent evaluation of aggregated behavioral metrics against predefined thresholds to detect abnormal health-related conditions. Primary alert generation is defined as

$$\text{Alert}_j = \begin{cases} 1, & D_j > \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

The severity of the inferred alert is computed as

$$S_j = \frac{D_j}{\tau}, \quad (15)$$

where $S_j > 1$ indicates an increasing level of inferred health risk requiring attention.

Here, D_j denotes an aggregated behavior deviation score derived from behavioral metrics.

B. Methodology

The proposed system adopts an edge-first intelligent vision pipeline for real-time cattle health monitoring, as shown in Fig. 1. Video streams acquired from multiple camera viewpoints are processed locally at the edge, where each frame undergoes preprocessing including resizing, region-of-interest extraction, and normalization. A YOLOv8-based deep learning model performs cattle detection, followed by multi-object tracking and OCR-assisted ear-tag identity association for continuous identity-aware monitoring. The framework supports concurrent multi-camera video stream processing through asynchronous inference pipelines, enabling stable low-latency edge deployment across multiple field-of-view regions. Cattle behavior recognition is performed using a CNN-based deep learning model with temporal modeling applied to smooth predictions and estimate behavior durations for continuous livestock health assessment.

C. Evaluation

The performance of the proposed edge-based monitoring framework is evaluated using real-world dairy farm video streams processed directly on the deployed edge device. The evaluation focuses on perception accuracy, identity reasoning reliability, behavior inference performance and real-time inference capability, consistent with the quantitative results reported in the Abstract.

1) *Detection Performance (mAP)*: The effectiveness of the learning-based cattle detection model is evaluated using mean

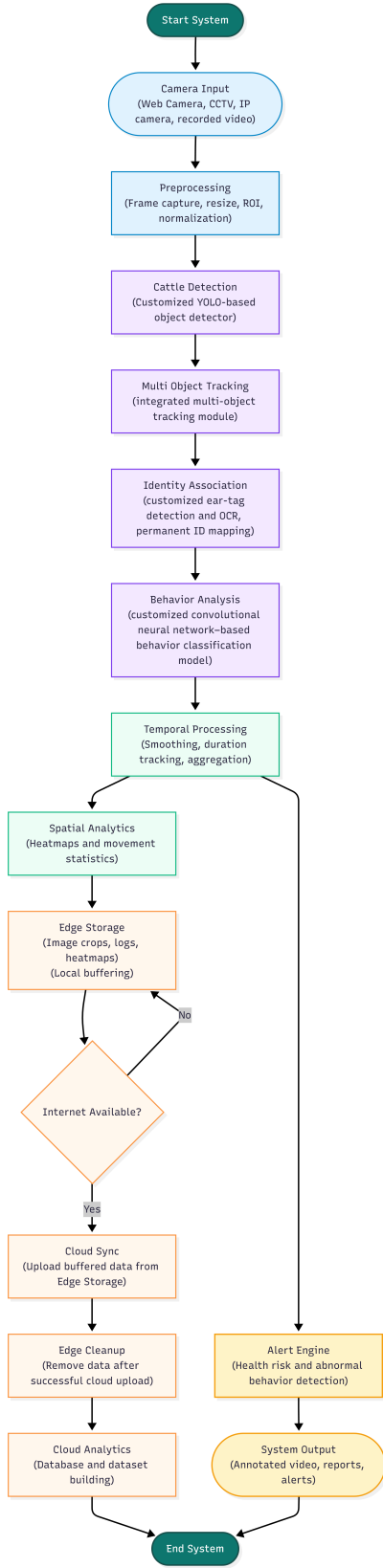


Fig. 1. Edge-based pipeline for vision-driven cattle detection, identity association and behavior inference.

Average Precision (mAP), computed over all detected cattle instances at a fixed Intersection-over-Union (IoU) threshold:

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^C \int_0^1 P_c(r) dr, \quad (16)$$

where C denotes the number of object classes and $P_c(r)$ represents the precision–recall curve for class c . This metric reflects the capability of the deep learning model to robustly localize cattle under real farm conditions. 2) *Identity Association Accuracy*: Identity association accuracy evaluates the reliability of the learning-assisted identity reasoning mechanism in maintaining consistent cattle identities across consecutive frames:

$$A_{\text{id}} = \frac{N_{\text{correct-id}}}{N_{\text{total-id}}}, \quad (17)$$

where $N_{\text{correct-id}}$ represents the number of frames in which cattle identities are correctly preserved and $N_{\text{total-id}}$ denotes the total number of identity assignments during tracking.

3) *Behaviour Classification Accuracy*: Overall behavior classification accuracy measures the correctness of the behavior inference model across all behavior categories:

$$A_{\text{beh}} = \frac{N_{\text{correct-beh}}}{N_{\text{total-beh}}}, \quad (18)$$

where $N_{\text{correct-beh}}$ denotes the number of correctly inferred behavior instances and $N_{\text{total-beh}}$ denotes the total number of evaluated samples.

4) *Macro-Averaged F1-Score*: To assess balanced learning performance across all behavior classes, particularly under class imbalance, the macro-averaged F1-score is computed as:

$$\text{F1}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \text{F1}_k, \quad (19)$$

where K denotes the number of behavior classes and F1_k represents the F1-score of the k -th behavior class. This metric ensures equal contribution from all behavior categories, including minority behaviors.

5) *Edge Real-Time Inference Capability*: The real-time inference ability of the framework deployed at the edge is evaluated based on the frames processed per second (FPS):

$$\text{FPS}_{\text{edge}} = \frac{N_{\text{processed-frames}}}{t_{\text{end}} - t_{\text{start}}}, \quad (20)$$

where $N_{\text{processed-frames}}$ denotes the number of frames processed during continuous operation and t_{start} and t_{end} represent the start and end times of the evaluation interval.

V. RESULTS AND DISCUSSION

A. Training Results

The deep learning-based cattle detection model achieved stable convergence and robust perception performance under practical farm conditions. As shown in Fig. 2, the precision, recall, F1-score and mAP metrics demonstrated smooth convergence behavior, indicating stable optimization of the proposed edge vision framework. Persistent cattle identity

inference through ear-tag detection and OCR-based association was also reliably maintained during continuous inference.

For behavior inference, the CNN-based behavior recognition model improved training accuracy from 83.7% to 96.2%, while validation accuracy consistently remained above 97%, demonstrating strong generalization capability. The confusion matrix in Fig. 3 indicates strong inter-class separability across the five behavior categories, with minor confusion primarily observed between standing and ruminating behaviors.

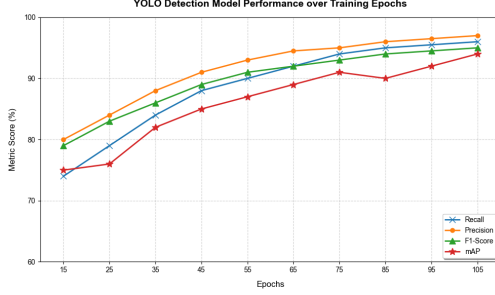


Fig. 2. Convergence behavior of the deep learning-based cattle detection model, showing precision, recall, F1-score and mAP across training epochs.

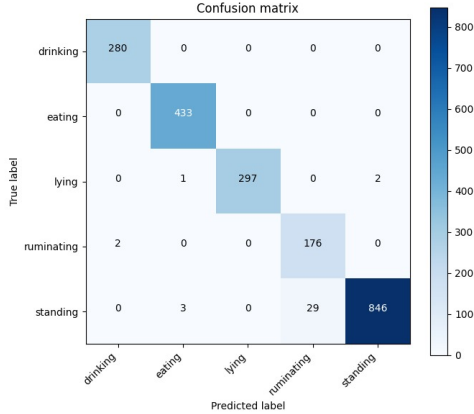


Fig. 3. Confusion matrix of the cattle behavior recognition model evaluated on the test dataset.

Table III summarizes the class-wise accuracy, precision and F1-score achieved by the standalone behavior recognition model on the test dataset containing 2,069 samples. The model achieved a macro-average F1-score of 96.71% and a weighted-average F1-score of 97.24%.

After integration with temporal smoothing, identity-aware tracking, OCR-assisted inference and multi-stage edge deployment optimization, the complete deployed framework achieved an overall behavior classification accuracy of 98.21% with a macro-average F1-score of 97.72%.

The strong classification performance is primarily attributed to ROI-based cattle extraction, temporal smoothing, class normalization during training and behavior-focused pre-processing, which together improve inference stability and reduce short-term prediction fluctuations during continuous real-time monitoring.

TABLE III
CLASS-WISE PERFORMANCE OF THE BEHAVIOR RECOGNITION MODEL

Behavior	Accuracy (%)	Precision (%)	F1-Score (%)
Drinking	98.74	97.92	98.11
Eating	98.43	97.85	98.02
Lying	98.27	97.64	97.91
Ruminating	96.84	89.42	92.08
Standing	97.91	98.36	97.42
Macro-Average	—	—	96.71
Weighted-Average	—	—	97.24

B. Field Trial Results

Field trials were conducted by deploying the complete edge-based vision framework within operational dairy farm environments. The system processed continuous multi-camera video streams in real time and executed deep learning-based cattle detection, multi-object tracking, persistent identity inference and behavior recognition directly on the edge device. The deployment environment presented realistic challenges, including illumination variation, partial occlusions, motion blur and dense cattle presence, allowing evaluation under realistic deployment conditions rather than controlled settings.

Fig. 4 demonstrates robust ear-tag detection and OCR-based identity inference under varying poses and occlusion levels, Fig. 5 illustrates real-time identity-aware behavior inference, where the framework associates the permanent ear-tag identity (Cow 085) with detected eating behavior across consecutive frames. Multi-cattle identity preservation and simultaneous behavior recognition are demonstrated in Fig. 6, while Fig. 7 presents qualitative behavior recognition results under practical farm conditions. Figures 8 and 9 illustrate the edge hardware setup and practical dairy farm deployment of the proposed monitoring framework. Additional qualitative results for drinking and ruminating behavior inference with confidence scores are presented in Fig. 10. Furthermore, Fig. 11 illustrates real-time Telegram-based alert generation for end-to-end monitoring and decision support.



Fig. 4. Qualitative results of learning-based ear-tag detection and OCR-based identity inference enabling persistent cattle identification.



Fig. 5. Real-time identity-aware behavior inference showing ear-tag-based identity assignment (Cow 085) and eating behavior recognition.



Fig. 6. Multi-cattle identity inference and standing behavior recognition enabled by the proposed edge vision framework.



Fig. 7. Field trial results illustrating computational intelligence-based eating, standing and lying behaviour recognition.



Fig. 8. Field deployment of the computational intelligence-driven edge vision system for real-time cattle behavior inference.



Fig. 9. Operational cattle shed deployment showing vision sensor placement and on-device computational intelligence execution for real-time cattle behavior inference under practical farm conditions.

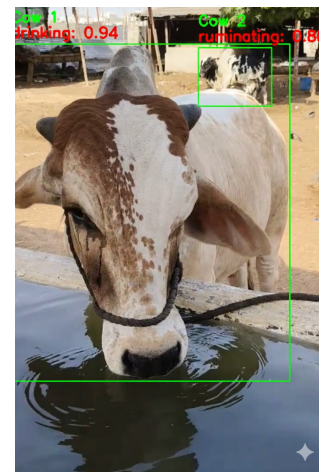


Fig. 10. Field trial results illustrating drinking and ruminating behavior inference with confidence scores.



Fig. 11. Real-time Telegram-based alerts generated by the edge vision framework for preventive cattle health monitoring.

C. Comparative Performance Analysis

Table IV presents a comparative analysis between the proposed framework and existing vision-based livestock monitoring approaches. Unlike several existing systems that primarily focus on isolated behaviour analysis or controlled experimental settings, the proposed framework integrates persistent cattle identity association, edge-based deployment and real-time behaviour inference within a unified intelligent monitoring pipeline. The comparison demonstrates the suitability of the proposed framework for practical real-time livestock monitoring under operational dairy farm environments.

TABLE IV
COMPARISON WITH EXISTING LIVESTOCK MONITORING APPROACHES

Reference	Accuracy (%)	Edge	Identity	FPS
Viazzi et al. [17]	89.3	No	No	Offline
Yu et al. [6]	93.2	Partial	No	10
Achour et al. [7]	94.1	No	Partial	Offline
Proposed Framework	98.21	Yes	Yes	14–18

D. Computational Cost and Edge Deployment Analysis

The proposed framework was deployed on the NVIDIA Jetson Orin Nano 8GB platform using CUDA-accelerated asynchronous inference for continuous real-time cattle monitoring. The YOLOv8 detection model processed multi-camera video streams at 640×480 resolution, while the EfficientNetV2-S behavior recognition model operated on cropped cattle regions resized to 384×384 pixels.

During field deployment, the framework achieved stable real-time inference performance of approximately 14–18 FPS under operational dairy farm environments involving illumination variation, partial occlusion, and dense multi-cattle scenes. Average GPU utilization remained between 70% and 82%, while CPU utilization ranged between 32% and 45%. The

complete inference pipeline consumed approximately 4.1–4.6 GB memory and operated within the 10–15 W embedded power profile of the Jetson Orin Nano platform.

The framework supports concurrent multi-camera video stream processing through asynchronous multi-threaded execution, enabling stable edge-level inference with reduced communication overhead and minimal network dependency. Experimental observations demonstrated that accurate cattle detection and stable tracking improved OCR reliability and behavior consistency, while temporal smoothing reduced short-term prediction fluctuations during continuous inference.

Although the framework demonstrated robust deployment performance, severe illumination variation, dense occlusion, and very low-light monitoring conditions may still affect detection accuracy and OCR reliability in certain scenarios.

E. Ablation Study

To analyze the contribution of major components within the proposed edge-based vision framework, an ablation study is performed by selectively disabling key modules and evaluating the resulting behavior recognition performance on the test dataset. The analysis focuses on the impact of temporal smoothing, OCR-based identity association and multi-object tracking on inference stability and overall classification robustness.

TABLE V
ABLATION STUDY OF FRAMEWORK COMPONENTS

Configuration	Acc. (%)	Macro F1 (%)
Without smoothing	95.84	94.91
Without OCR identity	96.27	95.42
Without tracking	94.76	93.88
Full framework	98.21	97.72

As summarized in Table V, disabling individual modules reduces the overall inference performance and behavioral consistency of the framework. Temporal smoothing improves prediction stability during real-time inference, while multi-object tracking enhances robustness under dense multi-cattle scenarios. OCR-based identity association supports reliable identity-aware monitoring and long-term behavioral aggregation. The complete integrated framework achieves the highest overall performance, validating the effectiveness of the proposed multi-stage edge deployment architecture.

VI. CONCLUSION AND FUTURE WORK

This work presented a low-cost, edge-based *computational intelligence* framework for real-time cattle behavior and health monitoring, validated under real-world dairy farm conditions. The proposed approach integrates learning-based cattle detection, multi-object tracking, ear-tag-driven identity inference and multi-class behavior recognition within a unified edge intelligence pipeline. The framework achieves 98.21% behavior classification accuracy with a macro-average F1-score of 97.72%, demonstrating suitability for deployment on resource-constrained edge hardware in smart agriculture environments.

In addition, future extensions of this work will explore the integration of multimodal sensing, including IoT-based wearable and environmental sensors, to enhance the accuracy and reliability of cattle health assessment. Further research will also aim to expand the behavior taxonomy and investigate adaptive and continuous learning strategies to improve the scalability and robustness of the proposed framework in dynamic farm environments.

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